Joint Positioning and Cooperative Wireless Sensing: Design and Over-The-Air Verification

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***Abstract*—The paradigm of wireless communications is shifting from data transfer to ubiquitous services. Perception of the surrounding environment has recently been highlighted as one of the key technologies enabling new applications, such as autonomous driving, extended reality, and healthcare. This movement gives a rise to an emerging concept of integrated sensing and communication (ISAC), where the distributed communication nodes, such as the base station and user devices, serve as probes for wide-area sensing, and the wideband radio signals are shared between communication and sensing. In recent years, a lot of effort has been directed towards the two main pillars of ISAC, i.e., self-localization of a node (positioning) and detection of device-free objects around the node (sensing). Nevertheless, only few of them have been implemented and brought up for real-world verification. This paper provides our latest efforts on validating the feasibility of wireless positioning and sensing. Through over-the-air measurement and processing in a 5G NR millimeter network with a commercial grade user equipment (UE), we demonstrate that the NR positioning and sensing techniques are mature enough to drive the evolution toward perceptive and cognitive wireless communications.**

I. Introduction

Evolution of wireless communications has consistently pursued enhancements, such as higher throughput, lower overhead, lower latency, and higher flexibility. These enhancements have motivated a paradigm shift of wireless communications from a data transfer service to new advanced services, including edge-computing, positioning, sensing, and AI/ML. Empowered by the new services, there have emerged innovative applications, such as autonomous driving, extended reality (XR), healthcare, and touchless user interface, which offer ubiquitous, immersive, and interactive user experiences that were not practically realizable earlier.

Another recent innovation, the Digital Twin (DT), has also found its new application in wireless communications and is gathering significant interest from both academia and industry [1], [2]. The DT represents a digitally created virtual counterpart of the real world. Particularly for wireless communications, the DT replicates the radio phenomena in the real world, such as reflection, refraction, diffraction, and scattering. By allowing computation or simulation on the digital proxy, the DT can help reducing the time, cost, and risk of real-world efforts, such as channel sounding [3], radio resource management [4], and network optimization/validation [5].

Hence, fidelity is a key requirement of the DT, and for some use cases, timeliness is another. To meet these requirements, maintaining synchrony between the DT and the real world is of paramount importance. As such, in recent studies, semi-real time environmental sensing with distributed sensors is perceived as an integral component of DT [6]. In wireless communications, in particular, each communication node can double as a sensor, and readily perform positioning and sensing measurement with radio signals, which then can be used to keep the DT up to date [7]. Along the same vein, 3rd Generation Partnership Project (3GPP) has noticed the importance of positioning and sensing technologies as a key enabler of sensing assisted communication and initiated specification work [8], which is a small but important first step toward the full-blown DT.

This paper is intended to demonstrate the latest status of industrial efforts on positioning and sensing in the 5G NR network, focusing on the mmWave (FR2) bands. In a NR FR2 network, a measurement campaign was conducted using commercial grade smartphones as the user-equipment (UE) devices. The collected Over-The-Air (OTA) data was first processed to compute UE position, and then for sensing the environments of the UEs. Through this effort, we prove the feasibility and maturity of the NR positioning and sensing technologies, and discuss the way forward for the next generation. This paper is a continuation of our previous work in [9], [10]. Hence, the main focus of this paper is on the enhancement and verification of the previously proposed algorithm.

The remainder of this paper is organized as follows. Section II gives an overview of the proposed joint positioning and sensing workflow. Section III offers details about the OTA test settings. Section IV analyzes the OTA test results, and concluding remarks are given in Section V.

II. Joint positioning and sensing workflow and algorithm

In this section, we provide a brief description the joint positioning and sensing workflow used in the OTA test, which is summarized as follows.

- Transmit/receive points (TRPs) transmit 5G NR Positioning Reference Signals (PRSs) on Transmission (Tx) beams
- UEs around the potential target objects measure Time of Arrival (ToA) and Angle of Departure (AoD) of Line of Sight (LOS) and reflected paths with the PRS signals.

1.1) ToA/AoD estimation from bistatic channel measurements with Tx and Receiving (Rx) beam sweeping

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1.2) ToA/AoD based NR positioning to determine the UEs' positions

1.3) ToA/AoD filtering to reduce reporting overhead

- Sensing server processes the ToA/AoD measurements and calculates the target positions.

2.1) Multi-object detection with RANdom SAmple Consensus (RANSAC) algorithm

2.2) Clustering and filtering detected objects

Some details including Multi-object detection algorithm can be found in [9], [10]. The major differences from our earlier method of [10] are as follows:

1 Estimated UE location provided by NR positioning is used instead of genie UE positions. Since the PRS is reused as the sensing reference signal, both positioning and sensing can be realized without additional OTA resource overhead. Nevertheless, some modifications are required for NR positioning as the Tx/Rx beam control logic is changed to suit the sensing scenario.

2 Monostatic sensing process is omitted in current procedures due to lack of full-duplex capability at the TRP. To build a bridge between the method in [10] and the OTA test environment, we assume that the sensing area, within which the potential target object is placed, is known and 4 out 32 PRS Tx beams from each TRP which are pointing towards the area are known at sensing server for the purpose of timing estimation.

3 AoD estimation is used to improve the sensing performance. Compared to Angle of Arrival (AoA) information adopted in [10], AoD has two major advantages:

i. Estimates of AoA or AoD are first computed relative to the antenna array, i.e., in a Local Coordinate System (LCS). To be usable for positioning or sensing, they need to be converted to angles in a Global Coordinate System (GCS), which needs knowledge of the array orientation. TRPs are typically stationary and their positions and panel orientations can be precisely measured and stored at the server in advance, while UEs' positions and orientations vary with time, requiring real time measurement which has lower accuracy.

ii. The array size and beamforming codebook design of the TRP lead to better angular resolution. UEs typically have fewer antenna elements due to form factor constraints, which leads to wider beams and thus to lower angular resolution of AoA estimates.

In step 1.2, a configuration of PRS is scheduled to cater to both AoD and Downlink Time Difference of Arrival (DL-TDoA) purposes. During each PRS occasion, a TRP is programmed to transmit M PRSs via distinct Tx beams. The scheduling of TRPs follows a round-robin approach. A UE selects one Rx beam to receive all the PRSs within a single occasion. This Rx beam changes in subsequent occasions until all N Rx beams have been used. The UE estimates both Reference Signal Received Power per Path (RSRPP) and the ToA of the LOS path for all the Tx/Rx beam pairs.

However, some of these measurements may be outliers, primarily due to misalignment between Tx and Rx beams. To address this, we employ a RSRPP-based pruning scheme for TDoA selection. For a pair of TRPs, denoted as TRP_i and TRP_j, there are a total of MN TDoA measurements and $2MN$ RSRPP measurements. Initially, we select the top K ToAs based on RSRPP from each TRP. We then identify the intersection of these two ToAs if their Tx/Rx beam pair matches. Subsequently, a median filter is applied to obtain the final TDoA estimation. This methodology ensures accurate and reliable measurements for DL-TDoA purposes.

The AoD estimation method in step 1.1 is a RSRPP-based interpolation method, similar to the method described in [11]. This is based on interpolation around the azimuth boresight angle of the Tx beam with the highest RSRPP among all the Tx beams as measured with the same Rx beam. The difference from [11] is that we pick the Rx beam based on the highest PRS RSRPP among Rx measurements, whereas in [11] the Rx beam is based on the serving Synchronization Signal Block (SSB). The AoD is then converted from LCS to GCS using the known orientation of the TRP panel. Note that the Tx beams of the TRPs used in the test only spread in azimuth directions, which limits our AoD estimation to azimuth angles.

After estimating the TDoA and AoD measurements, we formulate the three-dimensional positioning as a nonlinear optimization problem. The objective function is constructed to determine the estimate $\hat{\boldsymbol{u}}$ of the location of the UE as:

$$\hat{\boldsymbol{u}} = \underset{\boldsymbol{u}}{\operatorname{argmin}} \Big[\sum ((\|\boldsymbol{u} - \boldsymbol{s}_i\| - \|\boldsymbol{u} - \boldsymbol{s}_j\|) - c\tau_{ij})^2 + \sum w_i (g_i(2)/g_i(1) - \tan(\phi_i))^2 \Big] \tag{1}$$

where τ_{ij} denotes the TDoA between i-th and j-th TRP, and c is the speed of light. The estimated AoD from $\mathrm{TRP_i}$ is represented by ϕ_i.The UE and $\mathrm{TRP_i}$ locations are given by $\boldsymbol{u}$ and $\boldsymbol{s}_i$ respectively, which are 3-element vectors representing coordinates in the East-North-Up global Cartesian frame, and $g_i(m)$ is the m-th element in the vector $\boldsymbol{g}_i = \boldsymbol{u} - \boldsymbol{s}_i$. The weights for different AoD measurements are given by w_i. The two summations in (1) are over a subset of TDOAs τ_{ij} and AoDs ϕ_i respectively, and the subset is selected via a RANSAC based pruning algorithm.

A similar algorithm is used for ToA and AoD based sensing, where the objective function to determine the estimate $\hat{\boldsymbol{v}}$ of object position is formulated as:

$$\hat{\boldsymbol{v}} = \underset{\boldsymbol{v}}{\operatorname{argmin}} \Big[\sum ((\|\boldsymbol{v} - \boldsymbol{s}_i\| + \|\boldsymbol{v} - \boldsymbol{S}_k\|) - ct_{ik})^2 + w_{\mathrm{AoD}} \sum (\arctan(f_i(2)/f_i(1)) - \Phi_{ik})^2 \Big] \tag{2}$$

where t_{ik} is the estimated time of flight of the path from $\mathrm{TRP_i}$ to the k-th UE via the object. The AoD from $\mathrm{TRP_i}$ measured by k-th UE is represented by Φ_{ik}. The object and k-th UE locations are given by $\boldsymbol{v}$ and $\boldsymbol{S}_k$ respectively. $f_i(m)$ is the m-th element in the vector $\boldsymbol{f}_i = \boldsymbol{v} - \boldsymbol{s}_i$. Same AoD weight w_{AoD} is used for all TRPs here for simplicity. Similar to (1), the summations are over a subset of (TRP, UE) pairs that is selected by RANSAC. Other steps are the same as in [10].

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III. OTA TEST SETTINGS

In order to validate the proposed joint positioning and sensing design, we have established a testing environment within the premises of Qualcomm's Morehouse campus. The sensing target object was a cardboard box covered in aluminum foil. Four TRPs within the campus were set up to transmit PRS signals, which were received by a smartphone UE with Snapdragon® SM8650 chipset[1].

A. Test layout settings

Figure 1 shows an example of the UE on cart and the object we used in the test. The UE was placed on a 45 degree elevation tilted phone holder placed on a cart and its azimuth orientation was randomly selected. The target object was in the middle of lawn without any other object nearby.

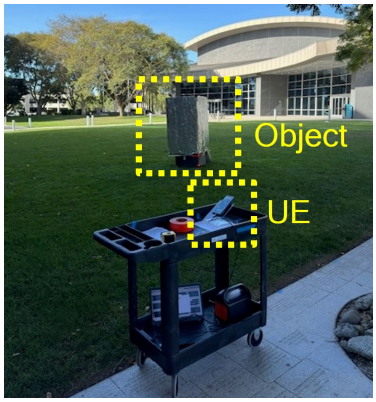


Fig. 1. UE on the cart and the sensing target object

The cart carrying the UE was relocated to 9 distinct positions surrounding the object, to collect PRS signals. This was done to simulate a situation where multiple UEs are concurrently sensing the object. In order to gain a comprehensive understanding of the performance of the proposed method, this test was replicated at 3 separate locations within the campus. These locations correspond to the three scenarios depicted in Figure 2.

B. RF settings

4 TRPs transmit a periodic PRS with 2 symbols and 64 resource blocks (RBs), resulting in a 100 MHz bandwidth with 120 kHz subcarrier spacing. The TRPs transmit 32 such PRS, conducting Tx beam-sweep across their codebook. The codebook contains 32 Tx beams with 3 degree spacing in azimuth, which covers approximately 120 degree area in front of the antenna panel. On the other hand, the UE possesses 3 panels, each with 27 Rx beams per polarization (each panel has dual-polarized antennas). Hence, the UE collects measurements for 126 different Rx beams. The TRPs and UEs are configured to execute beam sweeping in a sequential manner such that all possible Tx/Rx beam pairs will be measured once in one 'full beam sweep'. During the data collection session, we obtain 25 or more full beam sweeps

[1]Snapdragon and Qualcomm branded products are products of Qualcomm Technologies, Inc. and/or its subsidiaries.

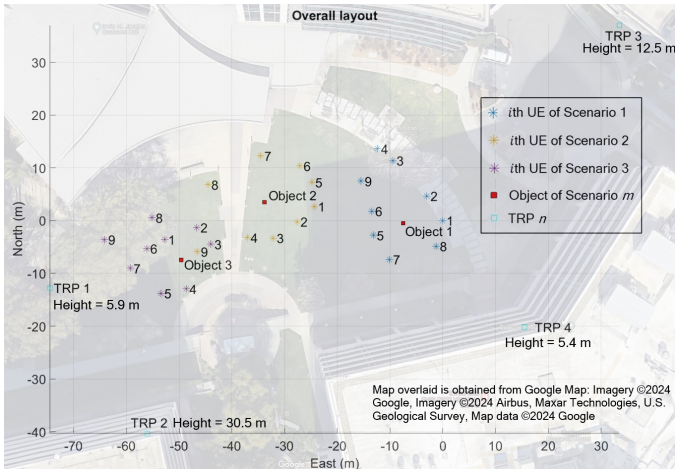


Fig. 2. The layout of 3 test scenarios

for each UE position in each of the 3 test scenarios. Note that one full beam sweep takes 4320 microseconds, which is short enough to assume the UE is stationary during the sweep.

IV. OTA TEST RESULTS ANALYSIS

A. Sensing based on genie UE location

We begin with an analysis of the sensing performance assuming perfect knowledge of UE positions (we will relax this assumption in the next section). Figure 3 illustrates an example of the sensing results for Scenario 3, utilizing only ToA information. The locations of detected objects across all beam sweeps in Scenario 3 are denoted by blue crosses. Note that within a single full beam sweep, our sensing algorithm may detect multiple objects, in which case we select one of them as the estimated position of the sensing target. The selection is based on nearest distance to the true geometric center of the object, and is marked with a red circle in Figure 3. Other estimated positions may correspond to other nearby objects (e.g., tree branches, light poles, or benches), or 'ghost' objects due to ground reflections, second-order reflections, or missed detection of LOS path.

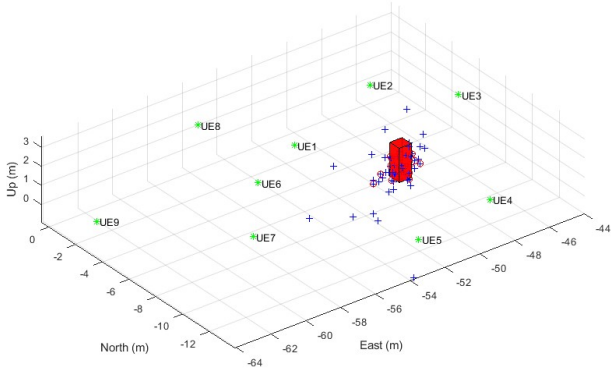


Fig. 3. An example of object detection results

The distance errors across all sessions of the three test scenarios are presented in Figure 4. We also highlight the average, median, and 80th percentile error of the three scenarios in Table I to facilitate a more direct comparison between different

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scenarios. From Figure 4, it is evident that in general, we can achieve a lower vertical error compared to the horizontal error in all three cases. Furthermore, an average 3D distance error of approximately 0.9 meters can be attained in the best case (Scenario 3). Note that in Scenario 3, the object is in close proximity to TRP_1, which may result in a strong reflection and consequently, superior performance. On the other hand, Scenario 2 exhibits the poorest performance, highlighting another crucial factor, namely the LOS condition and LOS path detection rate. At present, the time of flight of NLOS paths is estimated by calculating the excess delay relative to the path identified as the LOS path. If the LOS path does not exist (i.e., the LOS path is obstructed by another object) or it is too weak and thus submerged in the sidelobe of another NLOS path, we will derive an incorrect NLOS path delay, leading to errors in object detection. The LOS condition also affect the NR positioning accuracy, which will be shown in the next subsection.

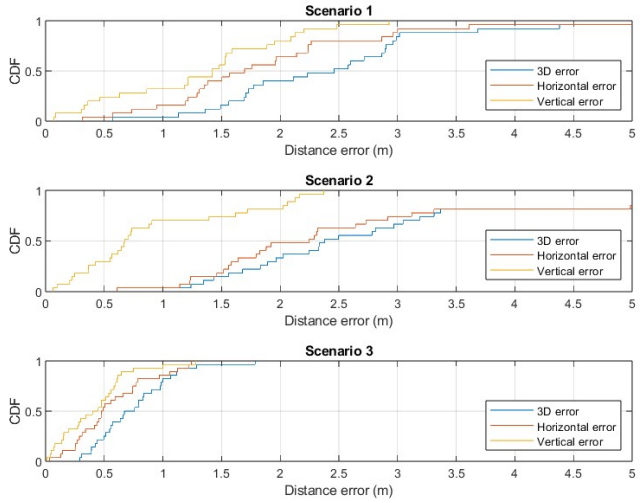


Fig. 4. Distance error distribution of object position estimation results with ToA information

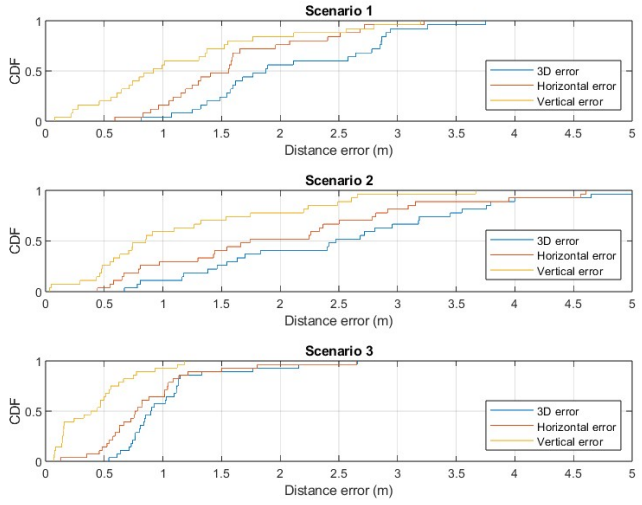


Fig. 5. Distance error distribution of object position estimation results with ToA and AoD information

To address the impact of AoD information, we have also evaluated the distance errors across all sessions of the three test scenarios using both ToA and AoD information in Figure 5. Upon comparing the results in Figures 4 and 5, it is evident that except in Scenario 3, the performance has improved, indicating that the AoD information can aid the algorithm. In general AoD is not guaranteed to help because it is possible that timing is much more accurate than AoD. Due to the long distance between TRPs and the object (around 50 meters or more), a small error in AoD could be translated to a large error in distance.

B. Sensing based on estimated UE location

In the preceding subsection, we assumed that the positions of the UEs are known with precision. However, this assumption may not hold true in the practical implementation of sensing. One potential method for determining the positions of the UEs, as mentioned before, could be NR positioning based on PRS.

1) NR positioning accuracy analysis: In Figure 6, we present the 3D positioning accuracy results utilizing both TDoA and AoD measurements. Our prior research in [11] achieved 2D positioning solely based on AoD estimates, assuming UE height is known. However, with the incorporation of additional TDoA measurements, we are able to accomplish 3D positioning with four or fewer TRPs.

The horizontal errors exhibit variability across different scenarios. The maximum error at the 80th percentile is observed in Scenario 3 and is less than 5 meters. Scenario 2 shows the poorest height estimation performance, while the errors in the other two scenarios slightly exceed 4 meters. It is important to note that the UE is used as the anchor for the sensing results. Consequently, the accuracy of its positioning has a direct impact on the sensing results. This underscores the significance of precise positioning in our study.

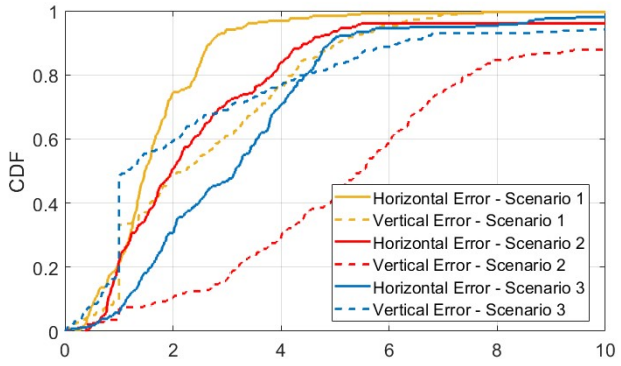


Fig. 6. Positioning estimation accuracy results

2) Joint positioning and sensing performance evaluation: In Figures 7 and 8, we demonstrate the distribution of distance errors in object location using estimated UE positions across the three scenarios. We also provide a summary of the 3D distance error in the 4 cases in Table I for easy comparison. Based on Table I and comparison of Figures 4 and 5 with Figures 7 and 8 respectively, we can see that a significant error is introduced when estimated UE positions are utilized. Specifically,

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for Scenario 2, the average error escalates to approximately 6 meters, due to the poor positioning accuracy depicted in the figure 6. While performance degradation is anticipated when introducing estimated UE positions, this degradation might be less pronounced if NR positioning accuracy significantly surpasses sensing accuracy (e.g., improving from the current 2 meter accuracy to sub-meter level accuracy). In this context, the accuracy of UE positioning emerges as a bottleneck for the entire joint positioning and sensing process.

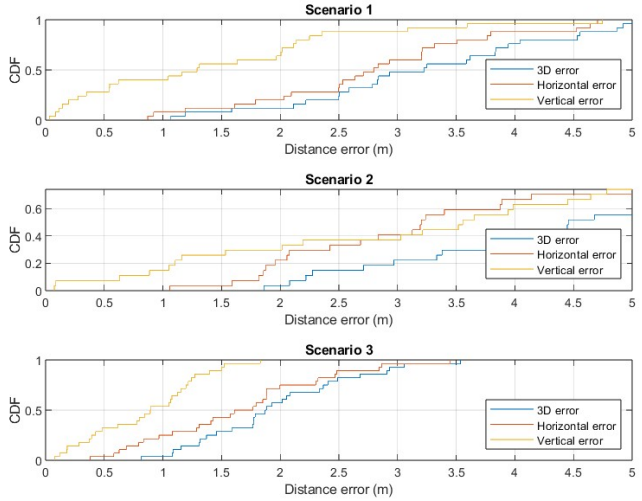


Fig. 7. Distance error distribution of object position estimation results with ToA information and estimated UEs' positions

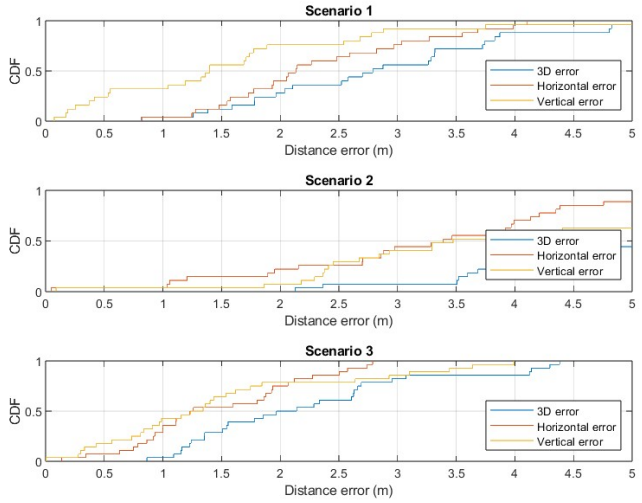


Fig. 8. Distance error distribution of object position estimation results with ToA and AoD information and estimated UEs' positions

V. Conclusion

In this paper, we demonstrate our efforts towards positioning and sensing in the 5G NR network in the mmWave band. We conducted a measurement campaign, utilizing commercial grade smartphones as UE devices, across 3 distinct areas. The OTA results reveal that our integrated positioning and sensing algorithm can achieve an average error distance of approximately 2 meters in good conditions. For continued research, one area of focus is on enhancing positioning accuracy, as it is a key factor influencing overall performance. For instance, the introduction of a positioning reference unit (PRU) could potentially mitigate positioning errors. Another avenue worth exploring is the Tx/Rx beam sweep strategy. The current approach of sweeping across the entire codebook results in substantial overhead. We believe these topics will be instrumental in further augmenting sensing performance and actualizing ISAC in real-world scenarios for 5G Advanced and 6G wireless networks.

TABLE I
Distance error summary for 4 cases

3D distance error (meter)		Scenario 1	Scenario2	Scenario 3
ToA only with Genie UE Pos.	Mean	2.44	3.12	0.76
	Median	2.46	2.38	0.72
	80%	3.02	3.37	0.98
ToA+AoD with Genie UE Pos.	Mean	2.10	2.49	1.04
	Median	1.88	2.48	0.90
	80%	2.90	3.55	1.13
ToA only with Estimated UE Pos.	Mean	3.28	5.82	1.96
	Median	3.23	4.46	1.88
	80%	4.56	9.88	2.41
ToA+AoD with Estimated UE Pos.	Mean	2.91	6.10	2.23
	Median	2.79	5.71	2.05
	80%	3.87	7.88	2.69

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